PERSONAL DATA ECOSYSTEM FOR LEARNING ANALYTI

Exploring the Impact of Lifestyle on Student Performance

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CSV

pandas

psycopg2

PostgreSQL

python

This project has been carried out as part of the module Introduction to Data Engineering of the Applied Data Science and Artificial Intelligence program at Zuyd University of Applied Sciences.

Involved organizations:

- Zuyd Applied Sciences and Artificial Intelligence.

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INTRODUCTION:

This project explores a student-centred system for securely collecting, managing, and connecting personal and educational data to support personalized learning through Learning Analytics and AI.

OBJECTIVE:

The objective of this project is to design and implement a secure and scalable relational data pipeline that integrates wearable, nutrition, and academic data from three students, enabling analysis of how sleep, physical activity, and diet relate to study habits and academic performance, while supporting personalized learning.

METHOD:

Data from WHOOP (physiological cycles, sleep, workouts), MyFitnessPal (nutrition logs), and institutional academic records (89 days, 3 students, 1,371 rows total) were extracted from CSV files, transformed, and loaded into a PostgreSQL relational database using a Python ETL pipeline (load_data.py). The schema comprises six linked tables with primary/foreign keys, CHECK constraints, and uniqueness constraints to ensure data integrity and idempotent loading. Five SQL queries were then designed to answer specific research sub-questions, each combining at least two data sources.

ETL PIPELINE

CSV Export › Extract + Clean › Transform + Validate › PostgreSQL Load › SQL Queries

DATABASE SCHEMA

whoop_cycles ‣ cycle_id (PK) → student_id (FK) recovery_score day_strain · hrv	**students** ‣ student_id (PK) S001 · S002 · S003 CASCADE DELETE (GDPR erasure)	**academic_perf.** ‣ record_id (PK) → student_id (FK) grade · study_hrs assessment_date
whoop_sleep ‣ sleep_id (PK) sleep_duration_hrs rem · sws · light	**whoop_workouts** ‣ workout_id (PK) sport · duration_min workout_strain	**nutrition_logs** ‣ log_id (PK) meal · calories_kcal protein · carbs · fat

RESULTS:

The relational pipeline successfully integrated 89 days of wearable, nutrition, and academic data from three students. Analysis revealed a clear pattern: higher physical activity, longer sleep, and better recovery were consistently associated with higher and more stable grades. S001, with the highest activity (36.4 min/day), sleep (~8.0h), and recovery (~78%), achieved grades between 7.3–9.6/10, while S003, with lower activity (8.2 min/day), sleep (~5.3 h), and recovery (~28–38%), obtained lower and more variable grades (3.6–7.1/10). S003 also showed below-average caloric intake before exams, coinciding with their lowest grades. However, with only three students, these findings indicate associations rather than causation.

Metric	Rodrigo (S001)	Rafael (S002)	Daniel (S003)
Avg sleep / night	7.8 hrs	7.1 hrs	6.3 hrs
Active min / day	36.4 min	22.0 min	8.2 min
Nutrition logged	89%	76%	43%
Avg study hrs / wk	18.2 hrs	14.5 hrs	9.8 hrs
Avg grade (0–10)	7.8	6.9	5.6

CONCLUSION:

This project demonstrates how a relational data pipeline can integrate personal and academic data to provide meaningful insights into student learning. The results suggest that sleep, physical activity, recovery, and nutrition may be related to academic performance and study habits. Although the small sample size prevents drawing causal conclusions, the project shows the potential of a student-centred personal data ecosystem to support personalized learning and future Learning Analytics applications.

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